

Problem

A balanced three-phase source with $V_L = 240$ V rms is supplying 8 kVA at 0.6 power factor lagging to two wye-connected parallel loads. If one load draws 3 kW at unity power factor, calculate the impedance per phase of the second load.

Step-by-step solution

Step 1 of 6

Calculate the value of phase angle of the source.

$$\cos \theta = \text{pf}$$

$$\cos \theta = 0.6$$

$$\theta = \cos^{-1}(0.6)$$

$$\theta = 53.13^\circ$$

Calculate the total real power of source.

$$P = S \cos \theta$$

$$= (8 \times 10^3)(0.6)$$

$$= 4.8 \text{ kW}$$

Step 2 of 6

Calculate the total reactive power of source.

$$Q = S \sin \theta$$

$$= (8 \times 10^3) \sin(53.13^\circ)$$

$$= (8 \times 10^3)(0.8)$$

$$= 6.4 \text{ kVAR}$$

Calculate the total complete power.

$$\mathbf{S} = P + jQ$$

$$= 4.8 + j6.4 \text{ kVA}$$

Step 3 of 6

Calculate the value of phase angle of load 1.

$$\cos \theta_1 = \text{pf}_1$$

$$\cos \theta_1 = 1$$

$$\theta_1 = \cos^{-1}(1)$$

$$\theta = 0^\circ$$

Calculate the real power of load 1.

$$P_1 = S_1 \cos \theta_1$$

$$= (3 \times 10^3)(1)$$

$$= 3 \text{ kW}$$

Step 4 of 6

Calculate the reactive power of load 1.

$$\begin{aligned}Q_1 &= S_1 \sin \theta_1 \\&= (3 \times 10^3) \sin(0^\circ) \\&= (3 \times 10^3)(0) \\&= 0 \text{ VAR}\end{aligned}$$

Calculate the complete power of load 1.

$$\begin{aligned}S_1 &= P_1 + jQ_1 \\&= 3 \text{ kVA}\end{aligned}$$

Step 5 of 6

Calculate the complete power of load 2.

$$\begin{aligned}S_1 &= S_1 + S_2 \\S_1 &= S - S_1 \\&= (4.8 + j6.4 \text{ kVA}) - (3 \text{ kVA}) \\&= 1.8 + j6.4 \text{ kVA}\end{aligned}$$

Calculate the expression for impedance per phase of the second load.

$$\begin{aligned}S_2 &= \frac{3V_p^2}{Z_p^*} \\S_2 &= \frac{3\left(\frac{V_L}{\sqrt{3}}\right)^2}{Z_p^*} \\S_2 &= \frac{V_L^2}{Z_p^*} \\Z_p^* &= \frac{V_L^2}{S_2}\end{aligned}$$

Step 6 of 6

Calculate the value of impedance per phase of the second load.

$$\begin{aligned}Z_p^* &= \frac{V_L^2}{S_2} \\&= \frac{(240)^2}{(1.8 + j6.4) \times 10^3} \\&= \frac{57.6 \times 10^3}{(1.8 + j6.4) \times 10^3} \\Z_p^* &= 2.346 - j8.34 \Omega\end{aligned}$$

Apply conjugate on both sides.

$$Z_p = 2.346 + j8.34 \Omega$$

Thus, the impedance per phase of the second load is $\boxed{2.346 + j8.34 \Omega}$.